



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Experiment 3

Water Jug Problem

Aim

To write a Python program to solve the Water Jug Problem using Breadth First Search (BFS).

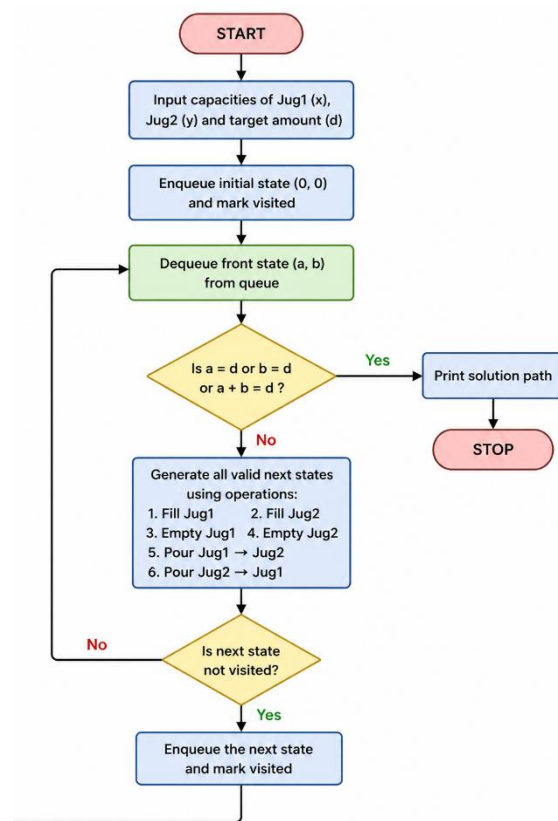
Objective

The objective of this experiment is to implement the Breadth First Search algorithm to solve the Water Jug Problem. It aims to find the sequence of operations required to obtain the desired amount of water.

Algorithm

1. Define the capacities of both jugs and the target amount of water.
2. Start from the initial state (0, 0).
3. Generate all possible next states using fill, empty, and pour operations.
4. Store each unvisited state in a queue.
5. Repeat the process until the target amount is reached.
6. Display the sequence of states leading to the solution.

Flowchart



Code

```
from collections import deque

def water_jug(x, y, target):
    visited = set()
    queue = deque([((0, 0), [])])
    while queue:
        (a, b), path = queue.popleft()
        if (a, b) in visited:
            continue
        visited.add((a, b))
        path = path + [(a, b)]
        if a == target or b == target:
```

```

        return path
    next_states = [
        (x, b),      # Fill Jug1
        (a, y),      # Fill Jug2
        (0, b),      # Empty Jug1
        (a, 0),      # Empty Jug2
    ]
    transfer = min(a, y - b)
    next_states.append((a - transfer, b + transfer))
    transfer = min(b, x - a)
    next_states.append((a + transfer, b - transfer))
    for state in next_states:
        if state not in visited:
            queue.append((state, path))

    return None

x = 4
y = 3
target = 2
solution = water_jug(x, y, target)
if solution:
    print("Solution:")
    for step in solution:
        print(step)
else:
    print("No solution found.")

```

Output

```
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>
===== RESTART: C:/Users/ABBIRAMY/Documents/AI-lab 3.py =====
Solution:
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
```

Result

The Python program successfully solved the Water Jug Problem and found the sequence of operations required to obtain the target amount of water.

Conclusion

The Breadth First Search algorithm efficiently explores all possible states and finds the shortest sequence of operations to reach the desired water quantity.